

Tier 3: Within-Patient Correlation (Repeated Measures)

GEE and Penalized Time-Varying Cox Models on the Hourly Panel

Abdul Ahad. Student ID: G00486649

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0.1 Overview

The longitudinal report (step 09) never modelled the within-patient correlation of repeated hourly measurements. This script addresses that gap in two classical ways: (1) GEE with an exchangeable working correlation, and (2) a penalized (ridge) time-varying-covariate Cox model on a counting-process formulation. Both operate on the sampled long panel (2,500 stays) for tractability.

0.2 Load Libraries and Set Seed

We load the GEE, sandwich, and survival packages, then fix the random seed for reproducibility.

```
suppressPackageStartupMessages({ library(geepack); library(sandwich); library(survival) })
set.seed(486649)
```

0.3 Locate and Read Data

A helper function searches several candidate paths for each input file. We read the sampled long panel that was produced by the earlier longitudinal pipeline.

```
find_data <- function(f) {
  cands <- c(file.path("../..//processed_data", f),
             file.path("processed_data", f), f)
  hit <- cands[file.exists(cands)][1]
  if (is.na(hit)) stop("cannot find ", f); hit
}
OUT <- dirname(find_data("panel_sample_long.csv"))
d0 <- read.csv(find_data("panel_sample_long.csv")) # full copy (for the Cox)
d <- d0
```

0.4 Define the Dynamic Incident Outcome

The stored per-hour label is nearly constant within a patient, so a random intercept would suffer complete separation. Instead we define a proper early-warning target on at-risk hours only: whether sepsis onset occurs within the next 12 hours. This outcome varies within septic patients and is the correct repeated-measures formulation.

```
# ---- DYNAMIC incident outcome on AT-RISK hours -----
# The stored per-hour `label` is patient-near-constant (all-0 for non-septic;
# 1 from onset-6h onward for septic) -> a random intercept would trivially
# separate it (complete separation). Instead we model the proper early-warning
# target on at-risk hours only: y = onset occurs within the next 12h.
# at-risk hour = non-septic (any hour) OR septic with hour < onset_hour
# This varies WITHIN a septic patient (0 early, 1 as onset nears) and is the
# correct repeated-measures outcome.
HORIZON <- 12
d <- d[is.na(d$onset_hour) | d$hour < d$onset_hour, ]
d$y <- as.integer(!is.na(d$onset_hour) &
                  (d$onset_hour - d$hour) <= HORIZON &
                  (d$onset_hour - d$hour) > 0)
cat(sprintf("Loaded sampled panel (at-risk hours): %d stay-hours, %d stays, incident-within-12h
            nrow(d), length(unique(d$stay_id)), mean(d$y)))
```

```
## Loaded sampled panel (at-risk hours): 76538 stay-hours, 2254 stays, incident-within-12h rate
```

0.5 Standardize Predictors

We median-impute missing values and z-score all predictors so that the resulting odds ratios are comparable across variables.

```
# ---- standardize predictors (median-impute; z-score for comparable ORs) ----
zt <- function(x) { x[is.na(x)] <- median(x, na.rm=TRUE); as.numeric(scale(x)) }
d$z_hr <- zt(d$hr); d$z_map <- zt(d$map); d$z_rr <- zt(d$resp_rate)
d$z_temp <- zt(d$temp_c); d$z_lac <- zt(d$lactate); d$z_hour <- zt(d$hour)
preds <- c("z_hr", "z_map", "z_rr", "z_temp", "z_lac", "z_hour")
form <- as.formula(paste("y ~", paste(preds, collapse=" + ")))
```

0.6 GEE vs Naive Pooled GLM

A per-patient random-intercept GLMM is degenerate for this outcome because most patients never have an event. GEE is the correct classical alternative: it estimates population-averaged effects with an exchangeable within-patient working correlation and cluster-robust sandwich standard errors. We contrast it with a naive GLM that treats the ~76k hours as independent, which substantially understates the standard errors.

```
# ---- (1) population-averaged GEE vs naive pooled GLM -----
# A per-patient random-intercept GLMM is DEGENERATE for this outcome: most
# patients never have an event, so their intercepts diverge (quasi-complete
# separation) and the variance component explodes. GEE is the correct classical
# alternative -- it models the population-averaged effect with an exchangeable
# WITHIN-PATIENT working correlation and cluster-robust ("sandwich") SEs, with
```

```

# no per-cluster parameters to blow up. Contrast with a naive GLM that treats
# the 76k hours as independent and so reports far-too-small SEs.
d <- d[order(d$stay_id, d$hour), ]
g_pool <- glm(form, data=d, family=binomial)

# (a) same GLM coefficients, but NAIVE (model-based) vs CLUSTER-ROBUST SEs.
# This isolates the effect of within-patient clustering on inference, holding
# the estimator fixed: treating 76k correlated hours as independent understates
# the SEs; the sandwich SE clustered by patient is the honest one.
se_naive <- sqrt(diag(vcov(g_pool)))
se_robust <- sqrt(diag(vcovCL(g_pool, cluster = d$stay_id, type = "HCO")))
# (b) GEE gives the interpretable within-patient working correlation.
cat("fitting GEE (exchangeable within-patient correlation) ...\n")

```

```
## fitting GEE (exchangeable within-patient correlation) ...
```

```

g_gee <- geeglm(form, id=stay_id, data=d, family=binomial, corstr="exchangeable")
alpha <- summary(g_gee)$corr[["Estimate"]]
cat(sprintf("exchangeable within-patient correlation alpha = %.3f\n", alpha))

```

```
## exchangeable within-patient correlation alpha = 0.676
```

```

nm <- names(coef(g_pool))
comp <- data.frame(
  term          = nm,
  OR_glm        = round(exp(coef(g_pool)), 3),
  SE_naive      = round(se_naive[nm], 3),
  SE_cluster_robust = round(se_robust[nm], 3))
comp$SE_inflation <- round(comp$SE_cluster_robust / comp$SE_naive, 2)
rownames(comp) <- NULL
cat(sprintf("\nCluster-robust vs naive SEs on the pooled GLM (within-patient alpha=%.3f):\n", al

```

```
##
```

```
## Cluster-robust vs naive SEs on the pooled GLM (within-patient alpha=0.676):
```

```
print(comp, row.names=FALSE)
```

```

##      term OR_glm SE_naive SE_cluster_robust SE_inflation
## (Intercept) 0.014  0.042          0.186          4.43
##      z_hr  1.206  0.021          0.051          2.43
##      z_map 0.880  0.021          0.047          2.24
##      z_rr  1.137  0.019          0.043          2.26
##      z_temp 1.083  0.020          0.046          2.30
##      z_lac  0.845  0.030          0.099          3.30
##      z_hour 0.175  0.043          0.193          4.49

```

```

write.csv(rbind(comp,
  data.frame(term="__alpha_within_patient__", OR_glm=round(alpha,4),
    SE_naive=NA, SE_cluster_robust=NA, SE_inflation=NA)),
  file.path(OUT, "tier3_gee.csv"), row.names=FALSE)

```

0.7 Penalized Time-Varying-Covariate Cox Model

We build a counting-process (start, stop] formulation of the hourly data and fit a ridge-penalized Cox model. This captures how time-varying physiology predicts the hazard of sepsis onset, with cluster-robust variance for within-patient correlation.

```
# ---- (2) penalized time-varying-covariate Cox (counting process) -----
# Build (start, stop, event] intervals per stay-hour; event at the onset hour.
# Uses the FULL sampled panel d0 (not the pre-onset-only GLMM frame).
cat("\nbuiding counting-process time-varying Cox (ridge) ...\n")

##
## building counting-process time-varying Cox (ridge) ...

dc <- d0[order(d0$stay_id, d0$hour), ]
dc$z_hr <- zt(dc$hr); dc$z_map <- zt(dc$map); dc$z_rr <- zt(dc$resp_rate)
dc$z_temp <- zt(dc$temp_c); dc$z_lac <- zt(dc$lactate)
dc$start <- dc$hour
dc$stop <- dc$hour + 1
# event = 1 at the onset hour of a septic stay, else 0
dc$ev <- as.integer(!is.na(dc$onset_hour) & dc$hour == dc$onset_hour)
# keep intervals up to and including onset (drop post-onset hours)
dc <- dc[is.na(dc$onset_hour) | dc$hour <= dc$onset_hour, ]
tv <- coxph(Surv(start, stop, ev) ~
            ridge(z_hr, z_map, z_rr, z_temp, z_lac, theta=1) + cluster(stay_id),
            data=dc)
cox_tab <- data.frame(
  term = c("z_hr", "z_map", "z_rr", "z_temp", "z_lac"),
  HR = round(exp(coef(tv)), 3))
cat("\nTime-varying Cox (ridge) hazard ratios for incident onset:\n")

##
## Time-varying Cox (ridge) hazard ratios for incident onset:
print(cox_tab, row.names=FALSE)

##      term      HR
##      z_hr 1.301
##      z_map 0.653
##      z_rr 1.064
##      z_temp 1.010
##      z_lac 0.891

write.csv(cox_tab, file.path(OUT, "tier3_tvcox.csv"), row.names=FALSE)
cat("\nTier 3 complete.\n")

##
## Tier 3 complete.
```